

Assignment 1

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Abstract

This report examines the detection and classification of various types of sexism on social media. While different Bidirectional LSTM models yielded similar results to one another, the Transformer-based RoBERTa model achieved superior performance. This improvement is likely due to RoBERTa's enhanced ability to analyze language within its specific context.

1 Introduction

Automated detection of sexist language is critical for social media safety. This report analyzes tweet classification into four categories (Direct, Reported, Judgemental, Non-sexist) comparing three architectures: a Bidirectional LSTM (BiLSTM), a stacked BiLSTM and a pre-trained Transformer (Twitter-RoBERTa). Experiments utilized multiple random seeds to ensure robustness. Results show that due to dataset imbalance, increasing BiLSTM depth yielded some gains (Stacked LSTM Macro-F1 $\approx$ 0.47 - Baseline LSTM Macro-F1 $\approx$ 0.36), whereas the Transformer outperformed them (Macro-F1 $\approx$ 0.53), confirming the superiority of pre-training for capturing linguistic nuance.

2 System description

2.1 BiLSTM Models

Data preprocessing: The text was filtered to alphanumeric characters using NLTK [1], replacing digits with a special `<num>` token. For normalization, we implemented context-aware lemmatization via the `WordNetLemmatizer` [2], using Part-of-Speech (POS) tags to reduce words to their syntactic roots.

Vocabulary definition: The vocabulary was constructed by merging tokens from the training set with pre-trained GloVe vectors. To handle Out-Of-Vocabulary (OOV) terms, embeddings were initialized as the mean of all known vectors, while a null vector represented the padding token.

Architecture: The Baseline and Stacked models employ: (i) an embedding layer initialized with pre-trained weights but left trainable to fine-tune representations, (ii) one and two Bidirectional LSTM layers respectively to ensure context learning in both directions, and (iii) a sigmoid-activated dense output layer for classification. The embedding layer was , employing masking (`mask_zero`) to ignore padding during calculations.

2.2 Transformer Model

We fine-tuned the `cardiffnlp's twitter-roberta-base-hate` checkpoint [5], leveraging its pre-training on 58 million tweets to handle social media noise. The pipeline utilizes BPE tokenization (truncated to 512 tokens) and `DataCollatorWithPadding` for dynamic batch padding to optimize efficiency. Training is performed via the Hugging Face `Trainer` API, wrapped in a custom evaluation loop across three random seeds to ensure robustness.

Implementation & contributions: While utilizing the standard RoBERTa architecture and `transformers` library, we adapted the classification head for our four-class task. Our primary contribution lies in the original experimental framework, specifically the logic for JSON data parsing, dynamic label mapping, hyperparameter grid search, and hierarchical model selection based on aggregated multi-seed performance.

3 Experimental setup and results

To optimize model performance, a grid search was conducted using three random seeds to ensure robustness, with the average Macro-F1 score on the validation set serving as the selection metric. For the BiLSTM models, in addition to standard parameters, we explored specific hyperparameters including LSTM units (64, 128, 256), embedding

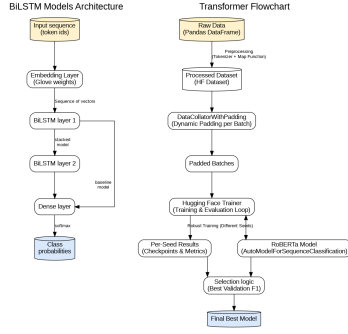


Figure 1: Transformer pipeline and BiLSTM architectures.

dimensions (50, 100), and class weighting strategies to address data imbalance, utilizing the Adam optimizer [3]. For the Transformer model, we examined various combinations of learning rates (1e-5, 2e-5, 3e-5, 2e-4), training batch sizes (8, 16), and number of epochs (3, 4, 5). The table below compares the mean Validation F1 ($\pm$ std across three seeds) with the Test F1 of the best selected model.

Model	Validation F1 (mean $\pm$ std)	Test F1 (best model)
Baseline Model	0.4736 $\pm$ 0.0328	0.3595
Stacked Model	0.4330 $\pm$ 0.0264	0.4730
Transformer	0.5951 $\pm$ 0.0293	0.5318

4 Discussion

4.1 Numerical Analysis

Experimental results validate that performance scales with architectural complexity. The Baseline LSTM showed limited capacity with 0.36 Macro-F1, whereas the Stacked variant improved context retention, raising Macro-F1 to 0.47 and notably enhancing Judgemental class F1 from 0.06 to 0.28. Ultimately, the RoBERTa Transformer achieved the best generalization with 76% accuracy, 0.53 Macro-F1 and 0.65 recall for Direct class, despite a slight performance drop on the rare Judgemental class due to data imbalance.

4.2 Error Analysis

We analyzed high-loss misclassifications to identify persistent model limitations. Detailed descriptions of Learning curves, Precision-Recall curves, and ROC curves are provided in the accompanying notebook. The analysis highlights primary error sources:

Tokenization and OOV: The impact of tokenization was explicitly visible. In LSTM models (static vocabulary), [UNK] tokens destroyed semantic context (e.g., masking “woke” in political critiques). The Transformer’s sub-word tokenization eliminated this, directly contributing to higher recall for explicit sexism.

Safe Opening Effect: Models often misclassify sexist tweets that begin with a neutral disclaimer followed by an adversarial conjunction (e.g., “*i dont care what gender... BUT...*”). The model overweights the safe introduction, ignoring the toxic conclusion.

Complex Register: Academic or sociological phrasing (e.g., “*violent antifeminism... vessel for breeding*”) is often mislabeled as Non-sexist due to the absence of standard slurs and the use of “high” vocabulary.

Qualitative analysis reveals distinct failure modes. While the Transformer resolved OOV issues, it still struggled with implicit sexism (e.g., sarcasm), frequently misclassifying nuanced examples as Non-sexist.

To address these issues, future work may implement **Focal Loss** to improve separability in latent space, **Back-Translation** to diversify training data for minority classes, or **Hierarchical Classification**.

5 Conclusion

The Transformer model outperformed BiLSTMs confirming the importance of attention in natural language classification tasks. However, distinguishing “Reported” and “Judgemental” sexism remains difficult, as this relies on deep context rather than just the presence of offensive words. Future work should focus on richer datasets, hierarchical classification and enhanced context handling.

References

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